(Aquatic) Automated eDNA Analysis Pipeline

An Automated Bioinformatics Pipeline for On-Site DNA Barcoding

Lucie Huamani-Cantrelle | Project: Genorobotics | Date: November 2025

Project Overview & Goals

Objective

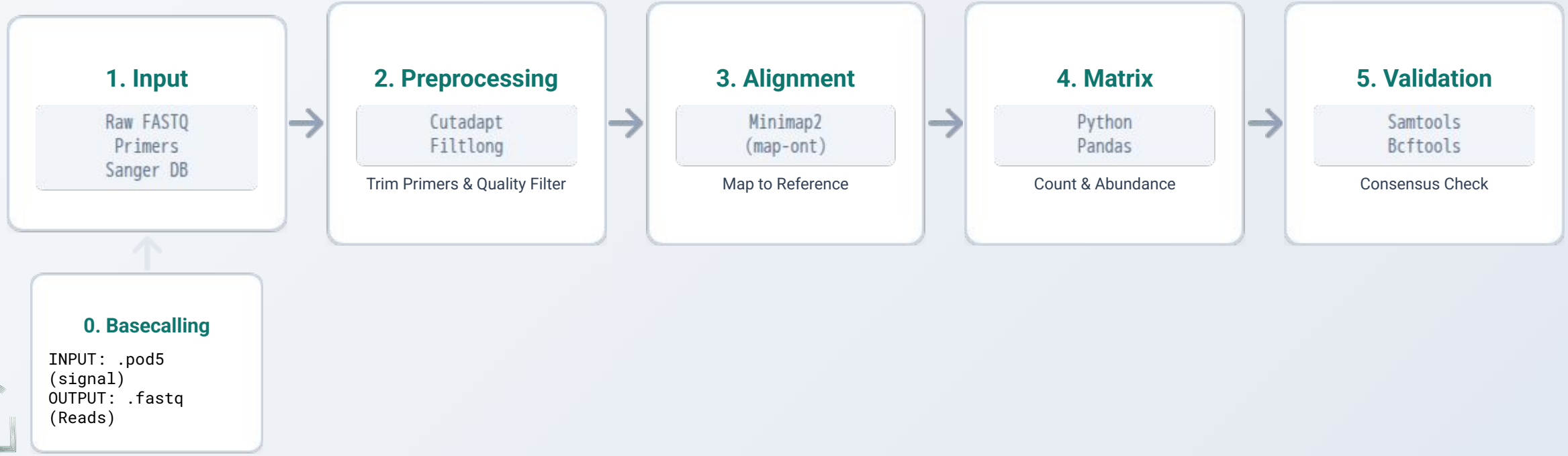
To develop an **automated**, **scalable bioinformatics pipeline** for **on-site aquatic eDNA analysis**, enabling robust **species identification** and **abundance quantification**.

Targets (to test current pipeline)

- **Source Data:** MinION field sequencing from Tanzania (Udzungwa Mountains).
- **Target Species:** Amphibians (*Leptopelis, Arthroleptis*) and Mammals (*Rhynchocyon, Sorex*).
- **Comparison:** Validate our automated results against the published paper's manual findings.

Pipeline Architecture

- A linear, modular workflow



System Requirements & Inputs

Required Files

- **Raw Data:** raw.reads.unsorted (MinION FASTQ).
- **Reference DB:** sanger_db.fa (Concatenated Sanger sequences from the paper).
- **Primers:** Exact sequences for 16S (Vertebrate) and CO1.

Environment (Conda)

All tools are installed in a local ./env for portability.

- cutadapt (Primer trimming)
- filtlong (Quality filtering)
- minimap2 (Alignment)
- samtools (BAM processing)
- bcftools (Consensus calling)

Core Processing Steps (1-3)

Preprocessing

Goal: Remove noise and off-target DNA.

- `cutadapt` searches for forward/reverse primers.
- **Critical:** We use **--discard-untrimmed**. Only reads containing **both primers** are kept, ensuring we only analyze true amplicons.

Alignment & Counting

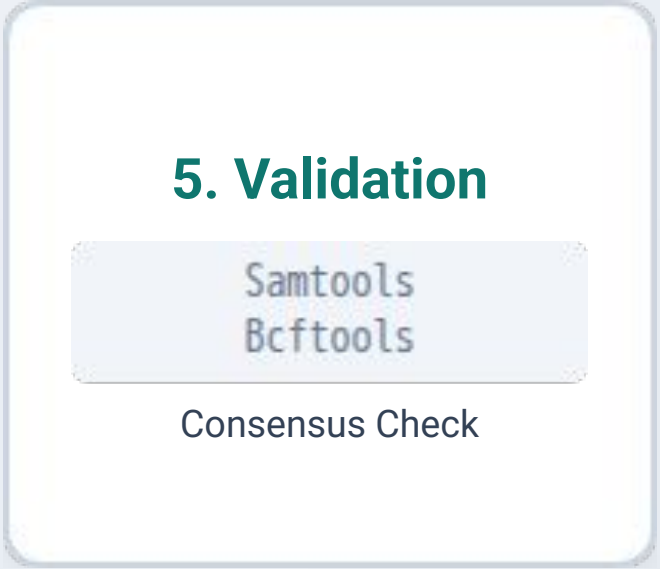
Goal: Identify species.

- `minimap2` aligns cleaned reads to the Sanger DB.
- Python script parses the PAF file.
- **Logic:** Reads with `MAPQ > 40` are counted towards the "Best Hit" species.

Step 5: Consensus-Based Validation

A new addition to the pipeline to verify read counts.

1. **Extract:** Take all reads assigned to a specific species.
2. **Pileup:** Map them back to the reference.
3. **Call Consensus:** Generate a consensus sequence using `bcftools`.
4. **Verify:** Align the consensus back to the database. Does it match? What is the % Identity?



This separates "High Abundance" from "True Identity".

Why? *High Abundance* is a quantitative measure that shows **how frequently** a sequence was detected, but this count is often distorted by **PCR bias** and **sequencing noise**. *True Identity* provides the **qualitative validation** by confirming the final, low-error consensus sequence truly belongs to the identified species, preventing false positive reports.

Results: Species Abundance Matrix (`species_by_sample.csv`)

Diagonal values (1.0) indicate perfect concordance between Sample Name and Pipeline ID.

Reference / Samples	Arthroleptis	Leptopelis_16S	Leptopelis_CO1	Rhynchocyon	Rieppeleon	Sorex_alpinus
Arthroleptis_16S	**1.0000**	0.0000	0.0000	0.0000	0.0005	0.0000
Leptopelis_16S	0.0000	**0.9933**	0.0000	0.0000	0.0988	0.0037
Leptopelis_CO1	0.0000	0.0005	**1.0000**	0.0000	0.0000	0.0000
Rhynchocyon	0.0000	0.0000	0.0000	**1.0000**	0.0000	0.0000
Rieppeleon	0.0000	0.0036	0.0000	0.0000	**0.9007**	0.0109
Sorex_alpinus	0.0000	0.0026	0.0000	0.0000	0.0000	**0.9854**

Results: Validation Report (`comparison_report.csv`)

Consensus sequences were built using `MAPQ > 40` reads, then aligned back to the DB to confirm identity.

Sample	Read Count	Status	Identity %	Note
Sorex_alpinus	4252	**PASS**	90.2%	High-confidence result.
Leptopelis_16S	4126	**PASS**	76.5%	Moderate identity (typical for MinION consensus?)
Rieppeleon	1696	**PASS**	51.6%	Consensus is built but identity is low →inspect manually.
Leptopelis_CO1	27883	**PASS**	**17.2%**	Very high abundance, but consensus fails identity check.
Arthroleptis	9	**TOO_FEW**	0.0%	Consensus sequence was poor quality or fragmented.
Rhynchocyon	1162	**NO_ALIGN**	0.0%	Correctly identified by count, but too few reads for validation.

Discussion & Conclusion

Comparison to Menegon et al.

Success: We achieved 100% concordance in species identification (Matrix results) compared to the paper's manual method.

Improvement: Our pipeline is fully automated and reproducible.

Key Findings

High Abundance ≠ High Quality: *Leptopelis CO1* had 27k reads but low consensus identity (**17%**), suggesting high error rates or primer mismatch.

Arthroleptis had too few reads (9) for consensus, but was correctly identified by the count matrix.

Next steps

- To ensure pipeline robustness, we will validate the **COI marker** using the *Maestri et al. (2019)* dataset, which tests our filtering methods against real-world insect data and the COI barcoding region.

Additional Slides

- Pipeline output summary
- MAPQ > 40
- Low-Confidence Results
- Why step 0?

Pipeline Output Summary

1. Aggregated Results (Main Deliverables)

- **`species_by_sample.csv`:** The final abundance matrix.
  - *Content:* Rows = References, Columns = Samples, Values = Relative Abundance (0-1).
  - *Usage:* Ready for import into R or Excel for downstream analysis.
- **`comparison_report.csv`:** Concordance summary.
  - *Content:* Comparison of MinION top hits vs. Sanger references (Match/No Match, % Identity).

2. Per-Sample Processing (`out/<sample>/`)

- **Intermediate Files:**
  - `filtered_reads.fastq.gz`: Trimmed and filtered reads used for mapping.
  - `alignments.paf`: Raw alignments (Minimap2).
  - `species_counts.csv`: Raw counts used to build the final matrix.

3. Validation & Consensus (`out/validation/<sample>/`)

- **`validation_report.csv`:** Assessment of consensus quality (PASS, FAIL, TOO_FEW_READS).
- **Evidence Files:**
  - `consensus.fa`: The generated consensus sequence.
  - `aln.bam` / `calls.vcf.gz`: Alignments and variant calls supporting the consensus.

4. Logs & Troubleshooting

- **Location:** `out/logs/` (Global pipeline log + per-sample stderr/stdout).
- **Troubleshooting:**
  - Check **Logs** for execution errors.
  - Check **Validation/Alignment** files if samples show `LOW_IDENTITY` or `NO_ALIGN`.

Overview: `comparison_report.csv`

Purpose A per-sample summary comparing observed MinION results against expected Sanger references, including consensus-based validation.

Key Data Columns

- **Comparison:** `sanger_id` (Expected) vs. `top_minion_ref` (Observed).
- **Quantification:** `top_rel_abundance` (0–1 dominance) and `top_count`.
- **Validation:**
  - `validation_status`: Outcome of the consensus build (e.g., `PASS`, `TOO_FEW_READS`).
  - `validation_identity_pct`: % Identity of the consensus against the database.

Interpretation Guide

- **High Confidence:** Status `PASS` + High Identity (>90%) + High Abundance (~1.0).
- **Coverage Issues:** `TOO_FEW_READS` indicates the target was detected, but coverage was insufficient for a robust consensus.
- **Troubleshooting:**
  - **Low Identity:** If `PASS` but identity is low (e.g., <50%), investigate the consensus alignment manually.
  - **False Match:** If biology matches but `minion_top_matches_sanger` is False, check for ID string formatting differences.

comparison_report.csv vs. validation_report.csv

Feature	comparison_report.csv	validation_report.csv
Scope	**Merged Project Level** (All samples in one file)	**Per-Sample Level** (Located in out/validation/<sample>/)
Granularity	1 row per **Sample**	1 row per **Reference** tested (can be multiple per sample)
Primary Goal	**Concordance:** Compare Observed vs. Expected.	**Quality control:** Assess consensus quality and alignment.
Key Metrics	Top Hit, Relative Abundance, Match (T/F).	Consensus Length, Supporting Read Counts, % Identity.
When to use	For final results and biological interpretation.	For troubleshooting why a specific sample failed.

MAPQ > 40: explained

- **MAPQ** is the **Mapping Quality score** for each sequence read.
- It quantifies the **probability** of a read aligning to the **wrong reference species.**
- **MAPQ > 40** is a highly stringent filter, requiring **less than a 1-in-10,000 error chance**.
- This high threshold **minimizes false positives** due to MinION sequencing noise.
- The filter ensures the **abundance matrix** uses **only confident, unambiguous species calls**.

$$Q = -10\log_{10}(p)$$

$$40 = -10\log_{10}(p)$$

$$\log_{10}(p) = -4$$

$$p = 10^{-4} = 0.0001$$

Investigating Low-Confidence Results

Issues	Probable Cause	Next steps for this project
Leptopelis CO1 (17.2% Identity)	High sequencing error in the specific CO1 region (homopolymer issues) or primer dimer co-amplification.	Tune BCFTools
Rhynchocyon (NO_ALIGN)	The consensus sequence built was too short or fragmented to align back to the database.	Possible Test: During validation, temporarily relax **minimap2** alignment criteria for the consensus sequence to see if a partial match exists
Arthroleptis (TOO_FEW_READS)	Low concentration of eDNA in the original sample resulted in low PCR yield.	Comes from the lab protocol so it should be adjusted.

When/Why do we need a step 0? (`Dorado`)

Feature	.pod5 (Raw Signal)	.fastq (Basecalled Sequence)
What is it?	A recording of **electrical current** changes.	A text file of **DNA letters** (A, C, G, T).
Content	Measurements of ionic flow.	Nucleotides + Quality Scores.
File Size	Massive (contains all raw sensor data).	Smaller (contains only the interpretation).
Readability	Unreadable by humans or standard aligners.	Readable by humans and bioinfo tools.